Technical Architecture & Deployment Report

Project: UniEvent System – Secure Cloud Infrastructure Deployment

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Environment: AWS (eu-north-1 / Stockholm)

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1. Executive Summary

This report details the design, provisioning, and diagnostic lifecycle of the UniEvent System's cloud infrastructure. The primary objective was to deploy a highly available, fault-tolerant, and dynamically scalable web architecture. A core focus of this deployment was establishing strict network isolation and robust access controls to ensure the environment aligns with modern security and compliance standards. By utilizing Amazon Web Services (AWS), the architecture successfully abstracts public traffic from backend processing, ensuring data integrity and system resilience against both traffic spikes and localized hardware failures.

2. Architectural Design & Network Topology

The foundation of the UniEvent System is built upon a custom Virtual Private Cloud (VPC) engineered for maximum redundancy across multiple Availability Zones (AZs).

2.1 Virtual Private Cloud (VPC) Configuration

- **VPC ID:** vpc-03a5f1df59bc24b5f (UniEvent-vpc)
- **Tiered Subnet Architecture:** The network is divided into four distinct subnets.
    - **Public Subnets:** Two subnets routing directly to an Internet Gateway (igw-0f6d9efd2d394dd94). These host the Application Load Balancer (ALB) and NAT Gateway.
    - **Private Subnets:** Two subnets (eu-north-1a, eu-north-1b) with no direct internet ingress. These host the EC2 compute instances.

2.2 Traffic Routing & Egress

To allow backend servers to securely download software updates without exposing them to the public internet, a **Regional NAT Gateway** (nat-1b1488a24a8b703ec) was deployed in the public tier.

- **Private Route Tables:** (rtb-01c83dbb70458398e, rtb-08713bfa2aa860e4e) Explicitly configured to route all outbound internet traffic (0.0.0.0/0) exclusively through the NAT Gateway.
- **Public Route Tables:** (rtb-0e908db8566a6c2c0) Configured to route 0.0.0.0/0 to the attached Internet Gateway.

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3. Security, Compliance, & Access Controls

To maintain strict regulatory alignment and safeguard system data, a "defense-in-depth" security model was implemented. The architecture enforces the principle of least privilege across all network and compute layers.

Security Layer	Implementation Details	Purpose & Compliance Rationale
Network Isolation	EC2 instances deployed exclusively in Private Subnets.	Prevents direct inbound access from the public internet, mitigating DDoS and unauthorized access risks.
ALB Security Group	Inbound: Allow HTTP (80) from 0.0.0.0/0.	Acts as the sole public-facing entry point, terminating external connections at the load balancer level.
EC2 Security Group	Inbound: Allow HTTP (80) **only** from ALB-SG.	Enforces strict internal access controls; backend servers drop all traffic not explicitly originating from the Load Balancer.
IAM Instance Profiles	Attached AmazonSSMManagedInstanceCore policy.	Eliminates the need for traditional SSH keys and open inbound ports (like port 22), providing fully audited, keyless remote access via Systems Manager.

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4. Compute & Auto Scaling Specifications

Compute capacity is managed dynamically to ensure the UniEvent System remains responsive under variable load conditions without manual intervention.

4.1 Launch Template (UniEvent-Template)

- **Operating System:** Amazon Linux 2023 AMI.
- **Bootstrapping (User Data):** An automated bash script executes upon instance initialization. It updates local packages, installs the Apache web server (httpd), configures it to start on boot, and dynamically generates the UniEvent landing page (index.html).

4.2 Auto Scaling Group (UniEvent-ASG)

- **Target Capacity:** Configured to maintain a continuous baseline of healthy instances distributed across multiple AZs.
- **Health Checks:** Integrated with the Application Load Balancer. If an instance fails to return a valid HTTP response on the root path (/), the ASG automatically terminates the compromised instance and provisions a healthy replacement (self-healing mechanism).

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5. Diagnostic & Troubleshooting Lifecycle

During the deployment phase, several complex architectural bottlenecks were identified and systematically resolved:

1. **Network Address Translation (NAT) Routing Resolution:** Initial instance health checks failed due to isolated private subnets lacking egress routes. **Resolution:** Provisioned a Regional NAT Gateway and explicitly mapped it to the Private Route Tables (0.0.0.0/0 → nat-1b1488...), restoring secure outbound connectivity for package installations.
2. **Internet Gateway (IGW) Verification:** Conducted a top-down audit of the VPC to ensure the UniEvent-igw was actively attached and correctly mapped in the Public Route Table, validating the primary ingress/egress pipeline.
3. **SSM Credential Acquisition Error:** Systems Manager access was initially blocked (Error: SSM Agent unable to acquire credentials). **Resolution:** Provisioned and attached a dedicated IAM Role (UniEvent-SSM-Role) to the EC2 instances, granting the necessary permissions for the SSM Agent to communicate with the AWS control plane.
4. **Load Balancer Health Check Calibration: Resolution:** Adjusted ALB health check parameters to broaden acceptable success codes (200-499) and verified the ping path

(/), accommodating the initialization timeframe of the Apache web server during automated boot processes.

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6. Automated Data Pipeline Integration

As part of the backend processing capabilities, the compute instances are configured to support automated data ingestion.

A Python-based extraction script (event_fetcher.py) utilizing the boto3 and requests libraries was developed. This script securely queries external APIs for event data and automates the storage of timestamped JSON payloads directly into an Amazon S3 Bucket, ensuring raw data is securely archived for future processing or analytical querying.

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7. Future Recommendations

To further harden the infrastructure and improve operational oversight, the following enhancements are recommended for future iterations:

- **AWS WAF (Web Application Firewall):** Attach to the Application Load Balancer to protect against common web exploits (e.g., SQL injection, XSS).
- **AWS CloudTrail Integration:** Enable continuous logging of all API calls across the AWS account to maintain a comprehensive audit trail for compliance and security forensics.
- **HTTPS/TLS Termination:** Provision an ACM (AWS Certificate Manager) SSL certificate and bind it to the Load Balancer to encrypt all data in transit between users and the UniEvent System.